Formal Benchmark Report v1

Knoema Engine Phase 20

24 deterministic runs

4 scenarios x 2 approaches x 3 local model profiles

Methodology

Runs use deterministic local scoring formulas.
No external LLM API or external framework is invoked.
Artifacts can be regenerated with runner.py.

Run Matrix

Scenario A: memory recall
Scenario B: relationship dynamics
Scenario C: narrative branching
Scenario D: scalability

Metric Definitions

Recall@k, token efficiency, scalability actions/sec, branches per 100 turns.
Composite score is used only for paired sign testing.

Scenario A

A memory_recall knoema: recall 0.789, efficiency 4.662, throughput 1141.6, branches 6.65
A memory_recall naive_llm: recall 0.500, efficiency 1.000, throughput 707.6, branches 4.95

Scenario B

B relationship_dynamics knoema: recall 0.786, efficiency 4.482, throughput 1053.8, branches 7.84

B relationship_dynamics naive_llm: recall 0.495, efficiency 1.000, throughput 631.1, branches 5.83

Scenario C

C narrative_branching knoema: recall 0.784, efficiency 4.429, throughput 1096.0, branches 11.40

C narrative_branching naive_llm: recall 0.492, efficiency 1.000, throughput 667.2, branches 8.48

Scenario D

D scalability knoema: recall 0.780, efficiency 4.280, throughput 608.9, branches 5.97

D scalability naive_llm: recall 0.487, efficiency 1.000, throughput 311.4, branches 4.44

Memory Recall Figure

Source figure: results/figures/memory_recall.svg

A memory_recall knoema: recall 0.789, efficiency 4.662, throughput 1141.6, branches 6.65

A memory_recall naive_llm: recall 0.500, efficiency 1.000, throughput 707.6, branches 4.95

B relationship_dynamics knoema: recall 0.786, efficiency 4.482, throughput 1053.8, branches 7.84

B relationship_dynamics naive_llm: recall 0.495, efficiency 1.000, throughput 631.1, branches 5.83

Token Efficiency Figure

Source figure: results/figures/token_efficiency.svg

C narrative_branching knoema: recall 0.784, efficiency 4.429, throughput 1096.0, branches 11.40

C narrative_branching naive_llm: recall 0.492, efficiency 1.000, throughput 667.2, branches 8.48

D scalability knoema: recall 0.780, efficiency 4.280, throughput 608.9, branches 5.97

D scalability naive_llm: recall 0.487, efficiency 1.000, throughput 311.4, branches 4.44

Scalability Figure

Source figure: results/figures/scalability.svg

Agent counts: 5, 10, 25, 50.

Narrative Branching Figure

Source figure: results/figures/branching.svg

Branching is normalized per 100 turns.

Significance

Paired sign-test p-value: 0.000244

All paired scenario/model profiles favor the Knoema memory policy.

Raw Artifacts

results/raw.jsonl has 24 rows.

results/summary.md contains aggregate tables.

results/figures contains 5 SVG figures.

Naive Baseline

Naive LLM receives full-history prompt context every turn.
This is intentionally transparent and token-heavy.

Mesa Reference

Mesa is documented as a capability boundary.
No Mesa performance numbers are claimed in this report.

Scale and Theory of Mind

Phase 42 folds the 500-agent metropolis run into the formal bundle.
Phase 43 adds persona opt-in theory-of-mind and a deterministic Sally-Anne harness.

Metric	Knoema 500-Agent	Concordia	Stanford
Status	local run	reference only	reference only
Scale	500 x 20 seeds	not measured here	25-agent paper
Latency	p95 65.1 ms	adapter run needed	paper only
Peak memory	max 436.0 MB	adapter run needed	paper only
Throughput	mean 1657.5 act/s	adapter run needed	paper only
Artifacts	JSONL + JSON + SVG	appendix needed	appendix needed
ToM surface	persona opt-in	no public opt-in API	no public opt-in API
Sally-Anne	20/20	not reported	not reported

External References

Concordia is external and not vendored. See baselines/concordia_reference.md.

Stanford reference is paper or code only. See baselines/stanford_reference.md.

Reproducibility

The report avoids wall-clock measurements in committed raw artifacts.
The same source inputs regenerate the same JSONL, summary, and SVG outputs.

Limitations and Next Steps

Metrics are deterministic proxies, not field trial results.

External baselines require separately executed adapters.

Next: equivalent external adapters for Concordia and Stanford-style scenarios.